

MRConsultation

Asynchronous Interaction Framework for
XR Teleconsultation

Asymmetric & Asynchronous Mixed Reality Teleconsultation Framework

Bending Spatial Computing Around the Human Workflow via
Decoupled Interaction Layers.

[CORE THESIS SUMMARY]

A

HARDWARE ASYMMETRY

Scales an advanced graphic rendering pipeline down to standalone consumer hardware (Meta Quest 3) on the patient side while flattening inputs for standard desktop displays.

B

TEMPORAL STATE

Decouples consultation timelines using an asynchronous store-and-forward architecture to entirely bypass live scheduling conflicts.

C

SYSTEM PARADIGM

Eliminates the structural "double-headset trap" and clinical network jitter by converting spatial authorship into local metadata layers.

The Socio-Clinical Problem & Operational Friction

STRUCTURAL DOMAIN: PATIENT-SIDE FRICTION

01

Cognitive Lock

"White-Coat Syndrome" triggers acute memory occlusion and high psychological anxiety during rigid, real-time 15-minute clinical windows.

02

Data Translation

Untrained patients lack the necessary clinical vocabulary or spatial health literacy to translate multi-dimensional visceral physical sensations into flat descriptive jargon.

STRUCTURAL DOMAIN: CLINICIAN-SIDE FRICTION

01

Workflow Realities

Surgeons and specialists operate in highly optimized, screen-centric environments under severe time pressure.

02

Hardware Rejection

Clinicians actively reject immersive headsets due to immediate physical strain, strict operating room sanitation protocols, and workplace workflow disruption.

[THE SYSTEM PARADOX]

"Current medical XR models strictly enforce synchronous, symmetric headset deployment—maximizing operational friction for the surgeon and creating heavy bandwidth bottlenecks for home networks."

Core Research Questions & Hypotheses

INVESTIGATION VECTOR // RQ1

Q1

Interaction Strategy

Can an asynchronous, asymmetric interaction framework deliver high spatial authoring agency to untrained patients while maintaining zero workflow disruption for the clinician?

INVESTIGATION VECTOR // RQ2

Q2

AI Data Viability

Can synthetic anatomical volumes generated via 3D foundation models (nv-generate-ctmr) preserve sufficient clinical legibility to support patient-guided symptom mapping?

[THE BASELINE HYPOTHESIS: H0]

"Separating spatial inputs into a decoupled, low-bandwidth metadata layer allows perfect clinical reconstruction on a 2D desktop browser, proving that immersive symmetry is unnecessary for effective remote spatial consultation."

Upstream Mesh Pipeline & Generative AI

01

Synthetic Gen

Engine: NVIDIA
nv-generate-ctmr (MAISI-v2
Latent Rectified Flow).

Output: Synthesizes
high-fidelity 3D liver anatomy
with tumor boundaries on
local 24GB VRAM limits.

02

Hybrid Seg

Strategy: Micro-to-macro
cascade.

Execution: Isolates primary
lesions with MAISI native
masks and extracts
surroundings via
TotalSegmentator to
eliminate "Context Collapse."

03

Mesh Isolation

Process: Imports raw
dual-mask NIfTI files into 3D
Slicer.

Execution: Runs automated
label isolation to convert
volumetric voxel groups into
distinct, unoptimized 3D
boundary shells.

04

Normalization

Optimization: Blender
QuadriFlow remeshing and
uniform decimation.

Execution: Rebuilds assets
into clean quad topology.
Secures a locked 90-120 FPS
on mobile XR chipsets to
entirely eliminate patient
motion sickness.

Technical Deep-Dive: Resolution-Aware Resampling

01

The Fixed-Resolution Bottleneck

Upstream Limitation: The nv-generate-ctmr foundation engine operates at a rigid internal resolution of 256^3 @ 1.5mm spacing. Directly extracting this voxel data via standard Marching Cubes produces severe, jagged staircase artifacts that trigger immediate patient motion sickness.

02

The Mathematical Solution

Spatial Optimization: Nearest-neighbor mask resampling mathematically scales voxel boundaries up to the target configuration ($512 \times 512 \times 256$, spacing [0.75, 0.75, 1.5]mm) at runtime, capturing full liver hepatectomy pathology while remaining strictly within local 24GB VRAM hardware limits.

03

The Geometric Refinement Pipeline

Mesh Normalization: Extracted meshes are routed into Blender for automated QuadriFlow remeshing and uniform decimation. This smooths boundaries into a uniform quad-topology, establishing a continuous mathematical surface that allows the raycasting script (AnnotationController.cs) to calculate accurate normal vectors for normal-mapped decal placement.

Unity Data Architecture & Registries

TIER 01

Master Registry Container [Class: `OrganDatabase.asset`]

Acts as the centralized system lookup engine. Deploys case-insensitive OrdinalIgnoreCase string matching to bind automated upstream segmentation labels directly to internal clinical databases.



TIER 02

Data Containers Layer [Class: `OrganData.cs ScriptableObjects`]

Instantiates 11 independent instances covering individual anatomical systems (Liver, Gallbladder, Portal Vein). Isolates medical content updates entirely from underlying graphics code.



TIER 03

Serialized UI Matrix [Struct: `OrganFunctionEntry[]`]

Serialized nested array structures that dynamically push localized icon configurations, clinical titles, and descriptions straight into the layout engine at execution.

Scene Bootstrap & Rendering Stability

LAUNCH COROUTINE // BOOTSTRAP TIMELINE

01

MRACTIVATOR LAUNCH

MRAsynchronousActivator.cs initializes asynchronously at primary scene start.

02

FOVEATED SUPPRESSION

Temporarily disables foveated rendering systems to prevent compositor load spikes.

02

ATW REPROJECTION CLEAR

Clears asynchronous time warp (ATW) reprojection lines to ensure pixel-perfect initialization.

STABILITY SHIELD 01 // [OVROverlayPrewarmer.cs]

Compositor Slot Allocation

Activates hidden overlays for 2 pre-warm frames to reserve compositor slots and kill the 5ms menu activation spike.

STABILITY SHIELD 02 // [CanvasStabilizer.cs]

Scale Anomaly Interception

Intercepts Meta SDK scale anomalies at runtime to lock and freeze world-space canvas viewport sizes.

STABILITY SHIELD 03 // [TMPAtlasPrewarmer.cs]

Font Texture Pre-Baking

Pre-bakes high-density font atlas textures to prevent mid-session runtime font-generation rendering lag.

Spatial Interaction Modes & Physics

01

State Swapping Engine

SphereGrabController.cs swaps between Macro Mode (**RayInteractable**) and Micro Mode (**DistanceHandGrabInteractable**) using MeshRenderer toggles to protect Interactor SDK component hierarchies.

02

Kinematic Telekinesis

RayPullBehavior.cs drives an accelerating telekinesis speed ramp in **FixedUpdate** via **Rigidbody.MovePosition** toward the hand tracking palm before handoff.

03

Collision Protection

MRUKPenetrationResolver.cs executes **Physics.ComputePenetration** against real room scene meshes to stop virtual organs from clipping into physical walls.

04

Safety Tooltips

DisplacementTooltipManager.cs tracks transform variance. Shifting assets past a 4cm individual or 12cm group threshold automatically triggers camera-billboarded reset tooltips.

2x2 Bento Grid Dashboard (Patient UI)

BENTO 01

Clinical Info Engine

Renders dynamic text data populated straight from the target **OrganData** ScriptableObject via **OrdinalIgnoreCase** case-insensitive lookup strings, separating medical content from graphics code.

BENTO 02

Annotation Workspace

Controls the **AnnotationController.cs** object pool of 500 URP Decal Projectors stamped at 4mm intervals with a 1mm depth offset to stop z-fighting visual artifacts on organ colliders.

BENTO 03

Voice Notes System

Manages audio capture via **Microphone.Start**, automatically trimming trailing silent frames to compress payloads. Freezes vector-based UI states cleanly via reflection hooks.

BENTO 04

Text Input Interface

Launches Meta's native virtual system keyboard on-screen using explicit **shouldHideMobileInput=false** configurations inside a standard world-space **TMP_InputField**.

UI ANIMATION ENGINE

ENGINE PIPELINE // [**BentoLayoutController.cs**]

Driven by state machines. Tweens **LayoutElement.preferredWidth** over a 0.35s curve, automatically collapsing adjacent unselected panels.

The Decoupled Asynchronous Data Payload

INPUT

The Input Paradigm

Core medical meshes remain strictly **Read-Only** and un-tampered with, completely bypassing FDA product liability.

Patient interactions are encapsulated entirely inside an independent text-based JSON payload named **AnnotationSession**.

PAYLOAD

Encapsulated JSON Schema

[AnnotationSession.json]

```
{
  "sessionId": "UUID_9081",
  "patientId": "PT_4421",
  "recipientId": "DOC_201",
  "annotationEntries": [
    {
      "organPath": "root/liver",
      "transformMatrix": [...],
      "inputType": "pain|text|audio",
      "maisi_label_id": 104
    }
  ]
}
```

ENDPOINT

Asymmetric Endpoint

Pushing this kilobyte-scale text package allows the clinician web portal to perfectly reconstruct, replay, and review the entire immersive session.

Via simple **ImportAnnotationsFromJson()** loops on a standard desktop monitor—requiring zero headset hardware on the physician side.